Proposition 8: fixes

Updated checklist | latest source rechecked on 11 October 2026

Items 1, 2, and 3 are resolved. The fixed-sample scope, divergence domain, and grid-spacing lemma are now correct. Items 4-14 remain open. The detailed written review has also been updated. No thesis files were changed.

Notation: $a = h_{min}$, $b = h_{max}$, $\delta = \delta_{ij}^{r,N}$, $\varepsilon = \varepsilon_N$. The empirical energy is written Q_r below.

1. State the fixed-sample scope. [RESOLVED]

The proposition specifies "fixed data $X_1,\ldots,X_n$", fixed positive endpoints, and a fixed grid exponent. The sample and its size stay fixed as N tends to infinity. This item is fulfilled.

Source: bandwidth-selection chapter, lines 765-772.

2. Correct the divergence domain. [RESOLVED]

The proposition defines $P_2 := P \cap L^2(\mathbb{R})$ and takes $D : P_2 \times P_2 \to \mathbb{R}_{\geq 0}$. This correctly restricts the finite L^2 divergence. This item is fulfilled.

Source: bandwidth-selection chapter, lines 766.

3. Repair the preceding grid-spacing lemma. [RESOLVED]

The lemma now explicitly has fixed positive endpoints. Its proof uses the correct powered grid identity, $c_N = O(\delta)$, the $o(N)$ assumption, and the $O(\delta^2)$ remainder. Since $\Delta_{r,N}$ is a constant multiple of $1/(N-1)$, $\delta \to 0$ and $c_N \to 0$ follow. The fixed upper endpoint justifies the final remainder bound. The introductory wording and stronger $O(1)$ comment are also correct. No further correction is needed for this item.

Source: bandwidth-selection chapter, lines 542-564.

4. Replace the integer-count calculation.

The bound is correct, but the displayed counting argument fails at integer endpoints. Use:

$$\#I_N(A) \leq 2\lfloor A/\varepsilon \rfloor + 1 \leq 2A/\varepsilon + 1.$$

Source: bandwidth-selection chapter, lines 778.

5. Correct the Fourier lower bound.

Use absolute values and integrate between min(h,s) and max(h,s). Replace the bandwidth/index typo. After squaring, the exponential is $\exp(-b^2\omega^2)$. A valid constant in $D(h,s) \geq c|u(s)-u(h)|^2$ is:

$$c = a^{4-2r}/(4\pi) \int_{\mathbb{R}} |P_n(\omega)|^2 \omega^4 \exp(-b^2\omega^2)\, d\omega > 0.$$

Source: bandwidth-selection chapter, lines 785-794.

6. Carry the tail normalization through every line.

The tail estimate concerns $\varepsilon T_N(A)$, not the unscaled $T_N(A)$. Fix the dropped or inconsistent ε factors. The final constant also loses a factor two:

$$\limsup_{N\to\infty} \varepsilon T_N(A) \leq 2\int_A^\infty \exp(-ct^2)\, dt \leq \exp(-cA^2)/(cA).$$

Source: bandwidth-selection chapter, lines 782-804.

7. Use absolute values in remainder estimates.

Divide by $|\tau_{N,j}|$ when it is nonzero. Bound the cross term in $|\rho_{N,j}|$ with $|\delta|$ times the two norms. For $j = i_N$, the remainders are zero; do not divide by zero.

Source: bandwidth-selection chapter, lines 812-824.

Proposition 8: fixes

Checklist continued | items 8-14 retained from the earlier review

8. Fill the first red gap by citing Taylor's theorem.

Set $u = h^r/r$ and let $F(u)$ be the Gaussian KDE at bandwidth $(ru)^{1/r}$. Check $F \in C^2$ in L^2 and $B = \sup\|F''(u)\|_2 < \infty$ on the fixed interval. A cited Taylor bound then gives:

$\|r_{N,j}\|_2 \le (B/2)|\delta|^2$; choose $a_N(A) = BA/(2\sqrt{\lambda_N}) \to 0$.

Source: bandwidth-selection chapter, lines 807-814. Taylor source: BYU Math 346, Theorem 6.6.9 and Corollary 6.6.14.

9. Fill the second red gap with a uniform estimate.

With $M = \sup\|F'(u)\|_2$, use the first remainder bound and $\lambda_N\delta^2 \le A^2$ on $I_N(A)$. This is the reason for the exponential error tending to zero; $\lambda_N/N^2 \to 0$ supplies the mesh condition instead.

$b_N(A) := A^2(Ma_N(A)+\tfrac{1}{2}a_N(A)^2) \to 0$;

$\sup_{j\in I_N(A)} \lambda_N|\rho_{N,j}| \le b_N(A)$, and $\sup_{j\in I_N(A)}|\eta_{N,j}| \le \exp(b_N(A))-1 \to 0$.

Source: bandwidth-selection chapter, lines 820-845.

10. Justify continuity and strict positivity of the energy.

Use the Fourier representation and Gaussian domination to establish continuity of the derivative in L^2. The integral below is positive because P_n is continuous and $P_n(0) = 1$. This also verifies that the limit's denominator is nonzero.

$Q_r(h) = h^{4-2r}/(4\pi) \int_{\mathbb{R}}|P_n(\omega)|^2\omega^4 \exp(-h^2\omega^2)\, d\omega > 0$.

Source: bandwidth-selection chapter, lines 358-360; 847.

11. Explain why the local window avoids the endpoints.

Since $h_{i_N} \to h_0$ lies in the interior, both transformed endpoint distances multiplied by $\sqrt{\lambda_N}$ tend to infinity. Hence $I_N(A)$ is not truncated for fixed A and large N. Also state $\varepsilon_N \to 0$ before the Riemann-sum step.

Source: bandwidth-selection chapter, lines 776; 847-859.

12. Replace the final equality by a squeeze argument.

For fixed A, the full row sum is not equal in the limit to the truncated integral. Put $L_A = \int_{-A}^{A} \exp(-t^2Q_r(h_0))\, dt$, use the bounds below, and only then let $A \to \infty$.

$L_A \le \liminf_N \varepsilon R_{i_N}^{(N)} \le \limsup_N \varepsilon R_{i_N}^{(N)} \le L_A + 2\int_A^{\infty} \exp(-ct^2)\, dt$.

Source: bandwidth-selection chapter, lines 861-867.

13. Make the remainder notation consistent.

Use $\delta_{ij}^{r,N}$ and $\tau_{N,j}$ throughout. Replace the inconsistent δ_N, $\delta_{N,j}$, τ_N, and $r_{N.j}$, and distinguish i_N from an undefined i.

Source: bandwidth-selection chapter, lines 785-834.

14. Qualify the interpretation after the proof.

The red row-sum interpretation is asymptotic at interior bandwidths. Use Q_r, or explain $\beta = 2(1-r)$. "Slow change" refers to $u = h^r/r$; the first-derivative functional measures squared speed. The Perron-vector concentration in the subsection heading still requires a separate argument.

Source: bandwidth-selection chapter, lines 714; 871-880.

All line references: /Users/iljasukhanov/MASTERARBEIT/Abschlussarbeit_2021/LaTeX/chapters/5_bandwidth_selection.tex. Original derivations: output/analysis/proposition_8_review.md.